

prove-TLA — Repair Loop & Corpus Pipeline

Companion to docs/RESUME_W4.md — how the verify/repair loop and the W4 cross-family corpus flywheel work, and where the numbers stand.

Generated 2026-07-20

North star: an LLM that emits guaranteed-correct TLA+ specs, where the guarantee is **structural** — it comes from the verify-until-correct loop (SANY + TLC + non-vacuity + repair), never from the model weights alone. No fine-tune ever makes an LLM always-right; the toolchain simply refuses to return anything that didn't pass. The model's only job is to make the loop *converge* — good first drafts and repairs, few iterations, high yield — and to produce specs that are *meaningful*, not merely valid.

1. The core loop: generate/repair → verify → gate

Every candidate spec, whether freshly generated from a natural-language scenario or repaired from a deliberately corrupted known-good spec, goes through the same four gates. A failure at any gate feeds the concrete SANY/TLC error plus a fault-localized fragment of the spec back into the next model call (up to 4 repair rounds) rather than a generic retry — this is the single biggest lever the harness has found for convergence.

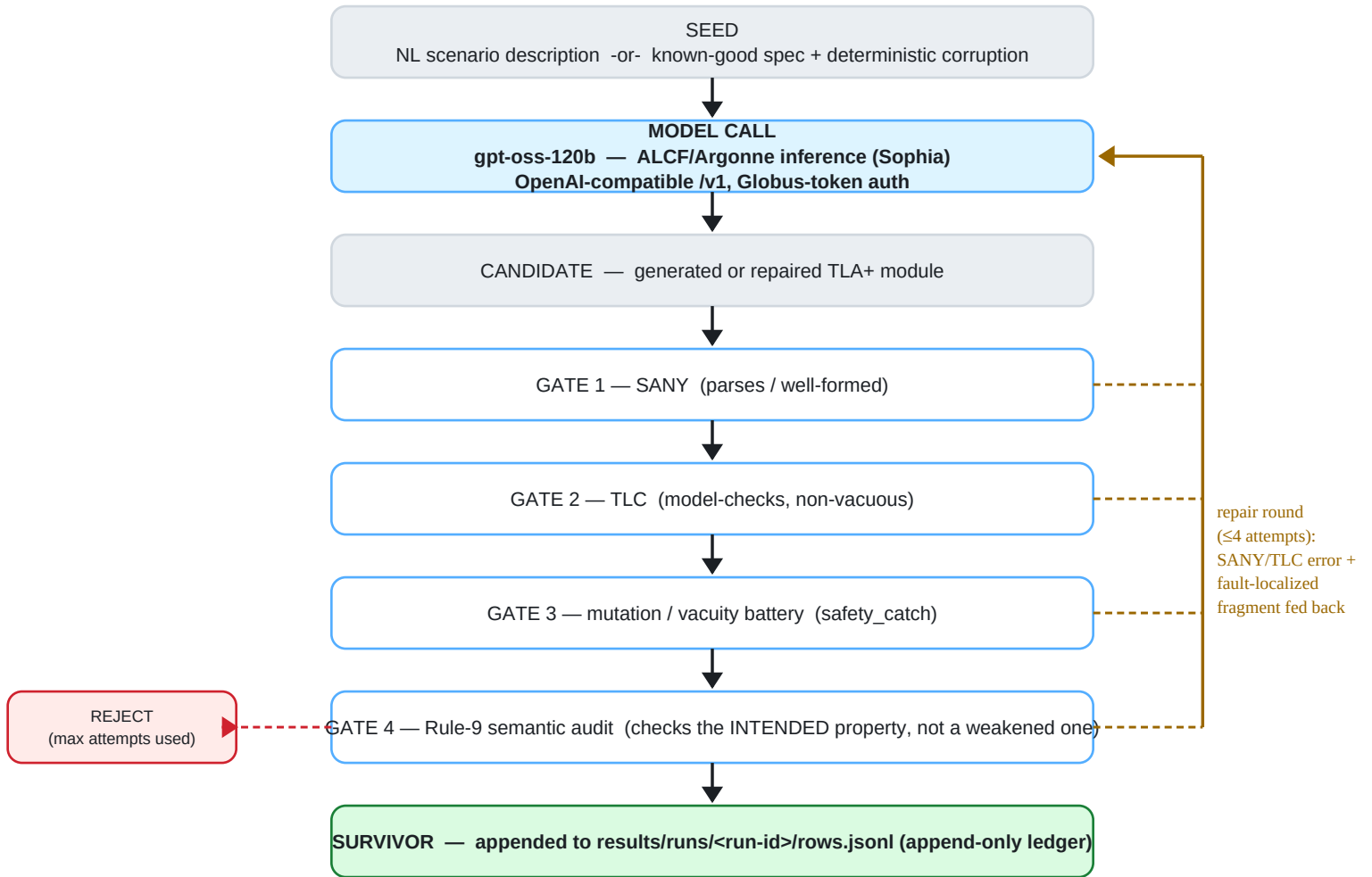


Figure 1 — the verify/repair loop (harness.repair, harness.gen_eval, harness.w4_verify_cell all instantiate this same shape).

2. Where the model calls actually run: ALCF / Argonne inference

All model calls (generation, repair, and the W2 self-family funnel) go through Argonne's ALCF inference API rather than a paid commercial API — this is free compute Eric has access to via an ANL identity.

Endpoint: `https://inference-api.alcf.anl.gov/resource_server/sophia/vllm/v1` — an OpenAI-compatible /chat/completions surface. Backing hardware is a node on Sophia, ALCF's DGX A100 cluster (8×A100-80GB/node).

Auth: a Globus refresh token, not a static API key — the harness shells out to

`~/venvs/alcf-inference/get_token.sh` per request (`OPENAI_API_KEY_CMD`) because tokens expire mid-sweep.

Model availability: only the **gpt-oss family** (120b, 20b) stays reliably hot; larger rotating models (Devstral, Llama-3.1-405B, Mixtral, Nemotron) cold-load or 408/503 for minutes. **20b was dropped from capability experiments on 2026-07-17** (Eric: "wouldnt trust a small model") — base **gpt-oss-120b** is the only arm that carries decision weight now; 20b is kept for cheap mechanics smoke only.

Training (as opposed to inference) is a separate, heavier path: real PBS jobs on Sophia/Polaris behind SSH + one-time-passcode MFA, currently gated — see §4.

3. The two corpus lanes that feed the flywheel

Every gate-passing spec is a candidate training row. Two independent lanes fill the same corpus, so that a future fine-tune isn't trained and re-tested on the model's own outputs alone (self-family rejection sampling was measured to cause a diversity collapse — see §5).

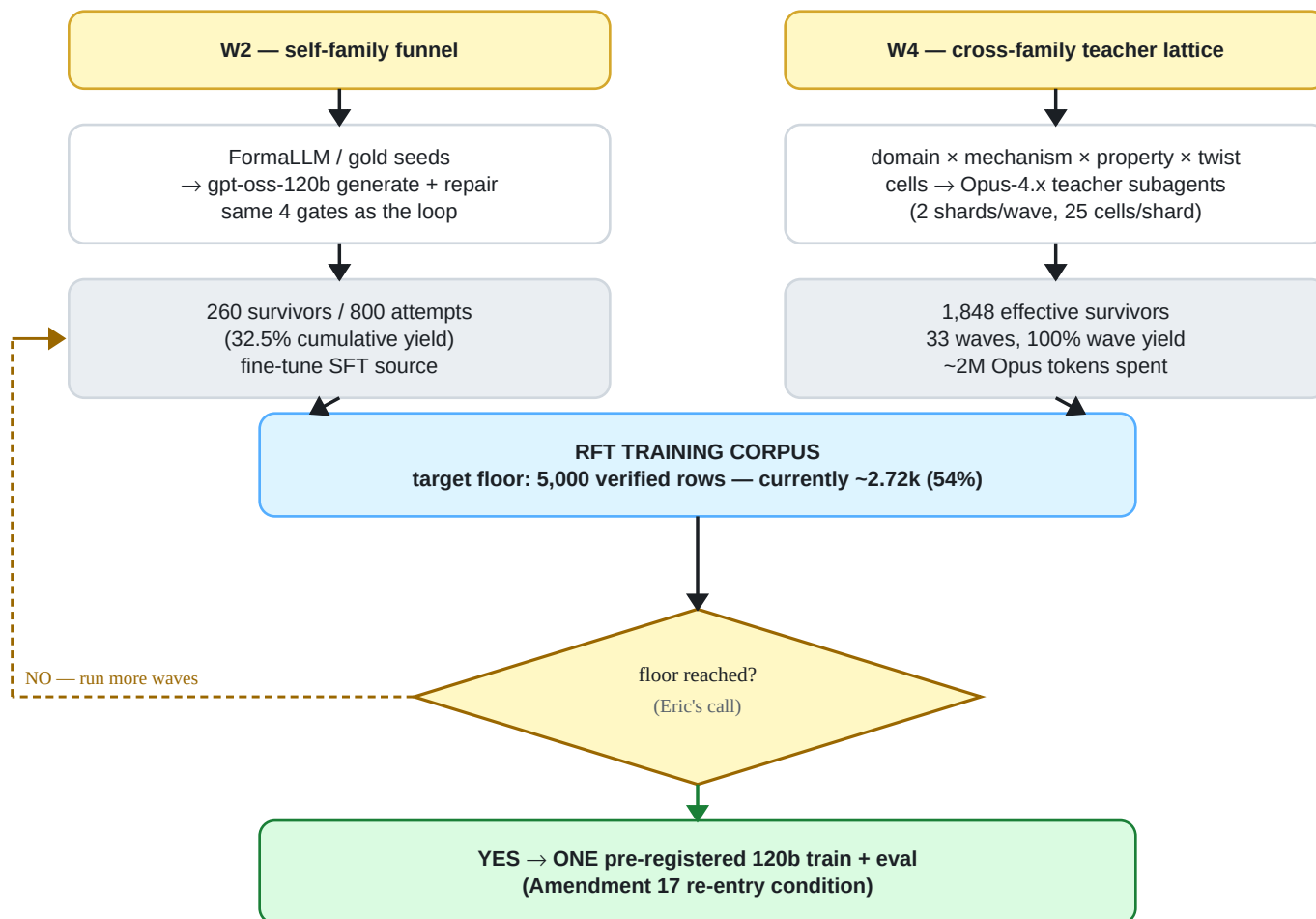


Figure 2 — W2 (self-family) + W4 (cross-family teacher) both fill the same RFT training-corpus floor, gating the next fine-tune decision.

4. Stage map (G1 corpus closure → G2 prover generalization)

Stage	What it measures	Status
Gate 0 — harness + oracle	Hand-verified positive closure of the 206-spec FormaLLM corpus (SANY + non-vacuous TLC)	SIGNED OFF — 170/206 (83%) frozen, hash-pinned
Gate 1 — Stage-1 repair sweep	Can the loop alone (no fine-tune) close more of the 206 via generate/repair?	SIGNED OFF — 176/206 (85%) oracle ∪ audited model
Gate 2 — fine-tune vs. frozen baseline	Does an RFT-fine-tuned 120b beat the base-model loop on the frozen 30-spec holdout (framings A/B)?	FAILED (Amendment 16, 2026-07-15) — A 11/30 vs 12/30, B 18/23 vs 21/23
Round 3 — loop-only capability	Structural scaffolding levers (prompt tier-1, grammar-constrained decoding tier-2) on base 120b	Both NULL (Amendment 19, 2026-07-18) — fine-tuning SHELVED, loop IS production
W4 — cross-family corpus (current)	Opus-as-teacher lattice corpus, building toward a 5k-row floor to re-earn a gated retrain	PAUSED 2026-07-19 — 1,848 survivors / 33 waves, ~2.72k/5k rows
Stage 3 — TLAPS prover (G2)	Unbounded proof discharge, not just bounded model-checking — deferred, separate deeper goal	Scaffolded — repair-and-extend signal 20/25, but 0/119 from-scratch proof-gen

5. Why fine-tuning is shelved right now

Two independent measurements (v2_sft2 generation-shaped, repair-v1 repair-shaped) both showed the same failure: pass@1 ticks up marginally but pass@4 **collapses** (17→9 in the repair arm) — the fine-tune's higher-temperature samples add zero new passing specs. That points at the **corpus source**, not the training recipe: self-family rejection sampling (the model grading its own generations) narrows the distribution instead of widening it. W4's cross-family teacher corpus (Opus-4.x generating structurally novel specs the funnel model never touched) exists specifically to break that loop — it's the one path back to a fine-tune worth re-running.

6. Where progress is right now (2026-07-20)

Metric	Value
Effective cross-family (W4) survivors	1,848, across 33 completed waves (100% wave-level yield)
Shards committed	0-73 (checkpoint commit 27ff3a)
Total verified corpus (W2 + W4)	~2.72k rows (~54% of the proposed 5,000-row train floor)
In-flight / stalled	Wave 34 (shards 74-75, cells 1850-1900) was dispatched but stopped before producing output — those cells are UNUSED and need re-assignment on resume
Cost per wave (measured)	~500k Opus subagent tokens, ~25-35 min wall-clock, audit catches ~1 incident per 2-3 waves (all recoverable so far)
Resume path	docs/RESUME_W4.md is session-agnostic: any Claude session can pick the loop back up from just that file + the repo (per-wave recipe, audit script, incident playbook, cost calibration)

Open decisions waiting on Eric

- Keep running Opus-teacher waves toward the 5,000-row floor, or call the corpus done earlier?
- Full-rate vs. half-rate wave cadence going forward.

- Trigger the ONE pre-registered 120b train + eval once the floor is hit (Amendment 17's re-entry condition).
- Publish the corpus to Hugging Face?

7. Integrity rules the loop enforces on itself

- **No self-certification** — Gate 0 closure is a frozen, hash-pinned positive enumeration, never inferred from model behavior.
- **Vacuity checking** — a TLC pass only counts once the mutation/trap battery has run on it.
- **Rule-9 semantic audit** — a pass counts only after confirming the spec checks the *intended* property, not a weakened or deleted one; this alone dropped one measured arm from a naive 170/206 to an honest 166/206.
- **Append-only ledgers** — every attempt is one JSONL row; corrections are new rows flagged keep-last, never edits to history.
- **W4-specific:** no mutation.py access by the generating agent (a shard that read it had its evidence flagged untrusted); complexity-tier LOC/var/action floors checked before any spec is appended; near-duplicate audit (shingle/Jaccard) runs after every wave.